

USING SINGLE-ROW FUNCTIONS TO CUSTOMIZE THE OUTPUT

1. Write a query to display the current date. Label the column Date.

```
select current_date from dual;
```

OUTPUT:

```
SQL> select current_date from dual;

CURRENT_D
-----
20-JUL-26
```

2. The HR department needs a report to display the employee number, last name, salary, and salary increased by 15.5% (expressed as a whole number) for each employee. Label the column New Salary.

```
select empno,ename,sal,sal + (sal * 15.5/100) as new_salary from emp;
```

OUTPUT:

```
SQL> select empno,ename,sal,sal + (sal * 15.5/100) as new_salary from emp;

  EMPNO  ENAME          SAL  NEW_SALARY
-----
   113  ADAMSSMITH      1100      1270.5
   101    KING          5000       5775
   102  SCOTLER        3000       3465
   103  TURNER         1500      1732.5
   104  SMITHKO         800       924
   105  BLAKE          2850     3291.75
   106  JONES          2975     3436.125
   107  ALLEN          1600      1848
   108  WARDEN          1250     1443.75
   109  MARTINKING      1250     1443.75
   110  JAMES           950     1097.25

  EMPNO  ENAME          SAL  NEW_SALARY
-----
   111  MILLERROBIN      1300     1501.5
   112  CLARKEN         2450     2829.75
   114  SARALA          2100     2425.5
   115  RAMESH          2600     3003

15 rows selected.
```

3. Modify the previous query to add a column alias that subtracts the old salary from the new salary. Label the column Increase.

```
SELECT empno,ename,sal,sal + (sal * 15.5/100) AS new_salary,
       (sal + (sal * 15.5/100)) - sal AS increase FROM emp;
```

OUTPUT:

```
SQL> SELECT empno,
           ename,
           sal,
           sal + (sal * 15.5/100) AS new_salary,
           (sal + (sal * 15.5/100)) - sal AS increase
        FROM emp;

  EMPNO  ENAME          SAL  NEW_SALARY  INCREASE
-----
   113  ADAMSSMITH      1100      1270.5      170.5
   101    KING          5000       5775        775
   102  SCOTLER        3000       3465        465
   103  TURNER         1500      1732.5      232.5
   104  SMITHKO         800       924         124
   105  BLAKE          2850     3291.75     441.75
   106  JONES          2975     3436.125    461.125
   107  ALLEN          1600      1848         248
   108  WARDEN          1250     1443.75     193.75
   109  MARTINKING      1250     1443.75     193.75
   110  JAMES           950     1097.25     147.25

  EMPNO  ENAME          SAL  NEW_SALARY  INCREASE
-----
   111  MILLERROBIN      1300     1501.5      201.5
   112  CLARKEN         2450     2829.75     379.75
   114  SARALA          2100     2425.5      325.5
   115  RAMESH          2600     3003         403

15 rows selected.
```

4. Write a query that displays the employee name (with the first letter uppercase and all other letters lowercase) and the length of the name for all employees whose last name starts with the letters J, A, or M. Give each column an appropriate label. Sort the results by the employees' names.

```

SELECT INITCAP(ename) AS employee_name,
       LENGTH(ename) AS name_length
FROM emp
WHERE ename LIKE 'J%'
      OR ename LIKE 'A%'
      OR ename LIKE 'M%'
ORDER BY ename;

```

OUTPUT:

```

SQL> SELECT INITCAP(ename) AS employee_name,
2       LENGTH(ename) AS name_length
3 FROM emp
4 WHERE ename LIKE 'J%'
5       OR ename LIKE 'A%'
6       OR ename LIKE 'M%'
7 ORDER BY ename;

EMPLOYEE_NAME          NAME_LENGTH
-----
Adamssmith              10
Allen                   5
James                   5
Jones                   5
Martinking              10
Millerrobin             11
6 rows selected.

```

5. Rewrite the query so that the user is prompted to enter a letter that starts the last name. For example, if the user enters H when prompted for a letter, then the output should show all employees whose last name starts with the letter H.

```
select * from emp where ename like 'H%';
```

OUTPUT:

```

SQL> select * from emp where ename like 'H%';

no rows selected

```

6. The HR department wants to find the length of employment for each employee. For each employee, display the name and calculate the number of months between today and the date on which the employee was hired. Label the column MONTHS_WORKED. Order your results by the number of months employed. Round the number of months up to the closest whole number.

```

SELECT ename,
       ROUND(MONTHS_BETWEEN(SYSDATE, hiredate)) AS months_worked FROM emp
ORDER BY months_worked;

```

OUTPUT:

```

SQL> SELECT ename,
2       ROUND(MONTHS_BETWEEN(SYSDATE, hiredate)) AS months_worked FROM emp
3 ORDER BY months_worked;

ENAME                      MONTHS_WORKED
-----
ADAMSSMITH                  470
MILLERROBIN                 534
KING                        536
JAMES                       536
RAMESH                      537
TURNER                      538
MARTINKING                  538
SARALA                      539
CLARKEN                     541
BLAKE                       543
SCOTLER                     543

ENAME                      MONTHS_WORKED
-----
JONES                       544
WARDEN                      545
ALLEN                       545
SMITHKO                     547
15 rows selected.

```

7. Create a report that produces the following for each employee:
 <employee last name> earns <salary> monthly but wants <3 times salary>.

```
SELECT ename || ' earns' || sal ||  
       'monthly but wants ' || (sal * 3) AS dream_salaries FROM emp;
```

OUTPUT:

```
SQL> SELECT ename || ' earns' || sal ||  
2         'monthly but wants ' || (sal * 3) AS dream_salaries FROM emp;  
-----  
DREAM_SALARIES  
-----  
ADAMSSMITH earns 1100 monthly but wants 3300  
KING earns 5000 monthly but wants 15000  
SCOTLER earns 3000 monthly but wants 9000  
TURNER earns 1500 monthly but wants 4500  
SMITHKO earns 800 monthly but wants 2400  
BLAKE earns 2850 monthly but wants 8550  
JONES earns 2975 monthly but wants 8925  
ALLEN earns 1600 monthly but wants 4800  
WARDEN earns 1250 monthly but wants 3750  
MARTINKING earns 1250 monthly but wants 3750  
JAMES earns 950 monthly but wants 2850  
  
DREAM_SALARIES  
-----  
MILLERROBIN earns 1300 monthly but wants 3900  
CLARKEN earns 2450 monthly but wants 7350  
SARALA earns 2100 monthly but wants 6300  
RAMESH earns 2600 monthly but wants 7800  
15 rows selected.
```

8. Create a query to display the last name and salary for all employees. Format the salary to be 15 characters long, left-padded with the \$ symbol. Label the column SALARY.

```
SELECT ename, LPAD(sal, 15, '$') AS salary FROM emp;
```

OUTPUT:

```
SQL> SELECT ename,  
2         LPAD(sal, 15, '$') AS salary  
3 FROM emp;  
-----  
ENAME  
SALARY  
-----  
ADAMSSMITH  
$$$$$$$$$$1100  
  
KING  
$$$$$$$$$$5000  
  
SCOTLER  
$$$$$$$$$$3000  
  
ENAME  
SALARY  
-----  
TURNER  
$$$$$$$$$$1500  
  
SMITHKO  
$$$$$$$$$$800  
  
BLAKE  
$$$$$$$$$$2850  
  
ENAME  
SALARY  
-----  
JONES  
$$$$$$$$$$2975  
  
ALLEN  
$$$$$$$$$$1600  
  
WARDEN  
$$$$$$$$$$1250  
  
ENAME  
SALARY  
-----  
MARTINKING  
$$$$$$$$$$1250  
  
JAMES  
$$$$$$$$$$950  
  
MILLERROBIN  
$$$$$$$$$$1300  
  
ENAME  
SALARY  
-----  
CLARKEN  
$$$$$$$$$$2450  
  
SARALA  
$$$$$$$$$$2100  
  
RAMESH  
$$$$$$$$$$2600  
15 rows selected.
```

9. Display each employee's last name, hire date, and salary review date, which is the first Monday after six months of service. Label the column REVIEW. Format the dates to appear in the format similar to: Monday, the Thirty-First of July, 2000

```
SELECT ename, hiredate, TO_CHAR(NEXT_DAY(ADD_MONTHS(hiredate, 6), 'MONDAY'),
                                'Day, "the" DDSP "of" Month, YYYY' ) AS review FROM emp;
```

OUTPUT:

```
SQL> SELECT ename,
2          hiredate,
3          TO_CHAR(
4            NEXT_DAY(ADD_MONTHS(hiredate, 6), 'MONDAY'),
5            'Day, "the" DDSP "of" Month, YYYY'
6          ) AS review
7 FROM emp;
```

ENAME	HIREDATE	REVIEW
ADAMSSMITH	23-MAY-87	Monday, the THIRTY of November, 1987
KING	17-NOV-81	Monday, the TWENTY-FOUR of May, 1982
SCOTLER	19-APR-81	Monday, the TWENTY-SIX of October, 1981
TURNER	08-SEP-81	Monday, the FIFTEEN of March, 1982
SMITHKO	17-DEC-80	Monday, the TWENTY-TWO of June, 1981
BLAKE	01-MAY-81	Monday, the TWO of November, 1981
JONES	02-APR-81	Monday, the FIVE of October, 1981
ALLEN	20-FEB-81	Monday, the TWENTY-FOUR of August, 1981
WARDEN	22-FEB-81	Monday, the TWENTY-FOUR of August, 1981
MARTINKING	28-SEP-81	Monday, the TWENTY-NINE of March, 1982
JAMES	03-DEC-81	Monday, the SEVEN of June, 1982
MILLERROBIN	23-JAN-82	Monday, the TWENTY-SIX of July, 1982
CLARKEN	09-JUN-81	Monday, the FOURTEEN of December, 1981
SARALA	15-AUG-81	Monday, the TWENTY-TWO of February, 1982
RAMESH	10-OCT-81	Monday, the TWELVE of April, 1982

15 rows selected.

10. Display the last name, hire date, and day of the week on which the employee started. Label the column DAY. Order the results by the day of the week, starting with Monday.

```
SELECT ename, hiredate, TO_CHAR(hiredate, 'Day') AS day FROM emp
ORDER BY TO_CHAR(hiredate, 'D');
```

OUTPUT:

```
SQL> SELECT ename,
2          hiredate,
3          TO_CHAR(hiredate, 'Day') AS day
4 FROM emp
5 ORDER BY TO_CHAR(hiredate, 'D');
```

ENAME	HIREDATE	DAY
SCOTLER	19-APR-81	Sunday
WARDEN	22-FEB-81	Sunday
MARTINKING	28-SEP-81	Monday
KING	17-NOV-81	Tuesday
TURNER	08-SEP-81	Tuesday
CLARKEN	09-JUN-81	Tuesday
SMITHKO	17-DEC-80	Wednesday
JAMES	03-DEC-81	Thursday
JONES	02-APR-81	Thursday
BLAKE	01-MAY-81	Friday
ALLEN	20-FEB-81	Friday
MILLERROBIN	23-JAN-82	Saturday
ADAMSSMITH	23-MAY-87	Saturday
SARALA	15-AUG-81	Saturday
RAMESH	10-OCT-81	Saturday

15 rows selected.

11. Create a query that displays the employees' last names and commission amounts. If an employee does not earn commission, show "No Commission." Label the column COMM.

```
SELECT ename, NVL(TO_CHAR(comm), 'No Commission') AS comm FROM emp;
```

OUTPUT:

```
SQL> SELECT ename,
2          NVL(TO_CHAR(comm), 'No Commission') AS comm
3          FROM emp;

ENAME                      COMM
-----
ADAMSSMITH                 No Commission
KING                       No Commission
SCOTLER                    No Commission
TURNER                     0
SMITHKO                    No Commission
BLAKE                      No Commission
JONES                      No Commission
ALLEN                      300
WARDEN                     500
MARTINKING                 1400
JAMES                      No Commission

ENAME                      COMM
-----
MILLERROBIN                No Commission
CLARKEN                    No Commission
SARALA                     No Commission
RAMESH                     No Commission

15 rows selected.
```

12. Create a query that displays the first eight characters of the employees' last names and indicates the amounts of their salaries with asterisks. Each asterisk signifies a thousand dollars. Sort the data in descending order of salary. Label the column EMPLOYEES_AND_THEIR_SALARIES.

```
SELECT SUBSTR(ename, 1, 8) AS employee,
       RPAD(' ', ROUND(sal/1000), '*') AS employees_and_their_salaries FROM emp
ORDER BY sal DESC;
```

OUTPUT:

```
SQL> SELECT SUBSTR(ename, 1, 8) AS employee,
2          RPAD(' ', ROUND(sal/1000), '*') AS employees_and_their_salaries
3          FROM emp
4          ORDER BY sal DESC;

EMPLOYEE
EMPLOYEES_AND_THEIR_SALARIES
-----
KING
SCOTLER
JONES
EMPLOYEE
EMPLOYEES_AND_THEIR_SALARIES
BLAKE
RAMESH
CLARKEN
EMPLOYEE
EMPLOYEES_AND_THEIR_SALARIES
SARALA
ALLEN
TURNER
EMPLOYEE
EMPLOYEES_AND_THEIR_SALARIES
MILLERRO
WARDEN
MARTINKI
EMPLOYEE
EMPLOYEES_AND_THEIR_SALARIES
ADAMSSMI
JAMES
SMITHKO

15 rows selected.
```

13. Using the DECODE function, write a query that displays the grade of all employees based on the value of the column JOB_ID, using the following data: PRESIDENT - A , MANAGER - B , SALESMAN - C , CLERK - D.

```
SELECT ename, job, DECODE(job, 'PRESIDENT', 'A', 'MANAGER', 'B', 'SALESMAN', 'C',
    'CLERK', 'D', 'No Grade') AS grade FROM emp;
```

OUTPUT:

```
SQL> SELECT ename,
2          job,
3          DECODE(job,
4                'PRESIDENT', 'A',
5                'MANAGER', 'B',
6                'SALESMAN', 'C',
7                'CLERK', 'D',
8                'No Grade') AS grade
9  FROM emp;
```

ENAME	JOB	GRADE
ADAMSSMITH	CLERK	D
KING	PRESIDENT	A
SCOTLER	ANALYST	No Grade
TURNER	SALESMAN	C
SMITHKO	CLERK	D
BLAKE	MANAGER	B
JONES	MANAGER	B
ALLEN	SALESMAN	C
WARDEN	SALESMAN	C
MARTINKING	SALESMAN	C
JAMES	CLERK	D
MILLERROBIN	CLERK	D
CLARKEN	MANAGER	B
SARALA	CLERK	D
RAMESH	ANALYST	No Grade

15 rows selected.

14. Rewrite the statement in the preceding exercise using the CASE syntax.

```
SELECT ename, job,
       CASE
         WHEN job='PRESIDENT' THEN 'A'
         WHEN job='MANAGER' THEN 'B'
         WHEN job='SALESMAN' THEN 'C'
         WHEN job='CLERK' THEN 'D'
         ELSE 'No Grade'
       END AS grade FROM emp;
```

OUTPUT:

```
SQL> SELECT ename,
2          job,
3          CASE
4            WHEN job='PRESIDENT' THEN 'A'
5            WHEN job='MANAGER' THEN 'B'
6            WHEN job='SALESMAN' THEN 'C'
7            WHEN job='CLERK' THEN 'D'
8            ELSE 'No Grade'
9          END AS grade
10 FROM emp;
```

ENAME	JOB	GRADE
ADAMSSMITH	CLERK	D
KING	PRESIDENT	A
SCOTLER	ANALYST	No Grade
TURNER	SALESMAN	C
SMITHKO	CLERK	D
BLAKE	MANAGER	B
JONES	MANAGER	B
ALLEN	SALESMAN	C
WARDEN	SALESMAN	C
MARTINKING	SALESMAN	C
JAMES	CLERK	D
MILLERROBIN	CLERK	D
CLARKEN	MANAGER	B
SARALA	CLERK	D
RAMESH	ANALYST	No Grade

15 rows selected.

